

Reducing Cloud Workload Costs in Geographically Distributed Data Centers with GeoSched

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Abstract—To serve millions of users around the globe with low-latency and high-availability services, prominent cloud service providers have placed their data centers across various locations globally. These data centers host diverse workloads, ranging from business-critical tasks such as web servers and email services to big data processing, real-time analytics, and high-performance computing. As the demand for cloud services continues to grow, it becomes crucial to have a workload provisioning mechanism that can minimize data center operating costs, minimize service level agreement (SLA) violations, and reduce user-perceived latency. In this regard, we propose GeoSched, a job provisioning framework that takes into account workload sensitivity, energy consumption, and cooling techniques used in the data center to schedule jobs across different locations. We evaluate our approach by conducting extensive simulations of actual scenarios and observe that GeoSched is superior to rival models in terms of energy efficiency. Specifically, our scheduler is capable of cutting down energy costs in data centers while meeting workload demands and maintaining SLAs. Since we acknowledge how SLAs are critical in cloud computing, our study recognizes the need for more descriptive SLAs to be included in our future work. These SLAs should be designed to capture the complex and diverse requirements of cloud workloads, such as response time, availability, throughput, and data consistency.

Keywords—Cost minimization, Cloud services, Data center, Cloud workstation

I. INTRODUCTION

Cloud computing enables various services to be provided over the Internet. These services include office applications, raw compute power, storage and a variety of Internet services including social media, multimedia streaming among many others. Rising online activity, be it for big data, digital media, or e-shopping boosts Internet traffic, thus necessitating a continuous effort to expand the capacity of data centers. This demand has increased the need for efficient data centers to handle the large amount computing resource requests especially among Internet companies like e Facebook, Google, Microsoft and Amazon. The size of these data centers can be more than 100,000 servers [1, 2] resulting in a huge costs, both to procure the IT infrastructure and to keep the systems running.

A typical data center consists of three major subsystems: servers and network cables form the IT equipments, power subsystem that supports the data center and the cooling equipments that maintain the data center's operating temperature at

levels where the IT components can work effectively. Running a data center requires a significant measure of energy, and in 2013, data centers in the United States consumed 91 billion kilowatt-hours worth of power [3, 4], which is double what the entire New York City demands. Much of the power goes into feeding cooling systems, the energy required to maintain a specific temperature in the data center. According to some reports, the cooling energy, which refers to the amount needed to maintain a stable temperature, can account for 50% [5] [6] [7] of a data center's energy transactions.

The aim of our paper is to introduce a meta-scheduler that can efficiently allocate jobs to data centers based on their cost-effectiveness. By doing so, this can help businesses with multiple data centers across different geographic locations to minimize their operational expenses. Our scheduling algorithm considers the costs associated with server operations and cooling expenses. The cooling costs are dependent on the type of equipment used, and in the case of conventional data centers, chillers are used to reduce temperatures of the hot air from CRAC (Computer Room Air Conditioning) units by means of mechanical refrigeration. This mechanism incurs large amount of power [8]. In recent times, many data centers have started using an economical method called air-side economizer for cooling their servers. This approach involves drawing in air from the outside into the data center while expelling the hot air stuck inside, thereby cooling the IT equipment. However, the use of air-side economizers is limited to regions with lower temperatures. We have recognized that the cooling efficiency of data centers varies based on their geographic location, which allows us to leverage this variability in temperature across geo-distributed data centers.

Let us describe this paper's structure. Section II showcases relevant studies on cost minimization of data centers and making cloud computing energy efficient. Section III narrates the workloads used in our experiments. Section IV gives details of the data center profiling. Section V defines the system model. Section VII provides details of the experiments and Section VI forms a reflective conclusion.

II. RELATED WORK

The topic of minimizing costs has been widely researched in both single data center and geo-distributed data center scenarios. In the past, efforts have primarily focused on reducing

the energy consumption of data centers in order to lower their operating costs. Gu *et al.* [9] studied how task assignment, data placement and data movement affected operating costs in data center networks, which spanned various geographical locations for big data applications. They employed a two-dimensional Markov chain to solve the model as a MILP problem. Similarly, Agarwal *et al.* [10] introduce Volley, an automated data placement mechanism designed for geo-distributed cloud services. The mechanism takes into account a range of factors such as WAN bandwidth cost, data center capacity limits, and data inter-dependencies to ensure optimal data placement across the network.

Yu *et al.* [11] presented an approach for minimizing energy costs in distributed Internet data centers (IDCs) that are integrated with smart microgrids. The proposed method takes into account various dynamic factors including variations in electricity costs, workload, renewable energy load, and power outage states. To solve this complex problem, the authors formulate it as a stochastic program and utilize the Lyapunov optimization technique. The work by Ren and He [12] introduce a novel online algorithm, named COCA, to minimize data center operational costs while aiming to move toward being carbon neutral. Unlike existing research in this area, COCA can manage resources at the distributed server-level, allowing each server to independently adjust its processing speed and determine the amount of workload to process in an optimal manner. It is important to note that this work is limited to a single data center with varying resource capacities.

A significant portion of the total energy consumption in a data center is attributed to cooling operations. As per certain reports, this can account for as much as 50% [5], [6], [7] of total energy chart. There has been a lot of work both at the application/middleware level [13], [14] as well as provisioning techniques [15], [16] that try to minimize the cooling energy.

Other research related to cooling efficiency include the work by Liu *et al.* [17] that enhances the sustainability of data center operations by minimizing electricity costs and reducing environmental impact. The approach accounts for several key factors, including renewable energy supply, dynamic pricing mechanisms, and cooling supply using chiller and outside air cooling. The authors first predict the renewable energy supply and IT demand and subsequently generate an IT workload management plan that schedules IT workload and allocates IT resources within the data center based on the time-varying power supply and cooling efficiency. This method is designed for a single data center environment.

III. CLOUD WORKLOADS

With cloud computing getting embraced by businesses worldwide, an increasing number of workloads are being migrated from traditional data centers to cloud-based solutions. Cloud data centers are designed to handle a wide variety of IT workloads, ranging from service-critical and latency-sensitive interactive applications, such as instant messengers and email services, to batch-style applications, such as scientific HPC and Big Data applications for financial analysis and Map Reduce processing. It is imperative to manage these workloads carefully to provide customers with low-latency services and to minimize how much energy goes into data centers responsible

Field Name	Type	Description
JobID	INT	Unique job identifier for each job
Arrival Time	INT	Arrival time in microseconds
Scheduled Time	INT	Arrival time in microseconds
Finish Time	INT	Arrival time in microseconds
Schedule class	INT	Latency class of the job.
Tasks	INT	Total number of tasks in the job.
Total CPU	INT	Aggregate sum of CPU used by all the tasks in the job
Total Memory	INT	Aggregate sum of memory used by all the tasks in the job
User	INT	Unique user identifier
SLA Param1	FLOAT	SLA parameter placeholder
SLA Param2	FLOAT	SLA parameter placeholder
SLA Param3	FLOAT	SLA parameter placeholder

TABLE I: Trace format description

for running these workloads. A good way to study these workloads are by analyzing real-world or synthetic workload traces. We look at cloud workload traces in detail in section III-A and III-C

A. Workload Traces

Unfortunately, applicability remains low in the work done by the cloud computing research community. There is limited work done to understand the diversity in cloud workloads. The first large-scale workload trace was released by Google in 2011 [18, 19], spanning 12,000 servers and 29 days. Other real-world traces include Wikipedia request trace [20] and Hadoop logs from Yahoo! [21]. There have also been attempts to generate synthetic workload traces using known real-world workload characteristics [22] [23]. For our experiments we use the workload trace from Google.

B. Custom Trace Format

To have a common trace format to represent different real-world and synthetic workload traces, we have formulated a trace format similar to standard workload format (swf) [24] defined for grid and supercomputing traces. Table I provides information about each field in the trace format. The format provides a uniform schema that can be used by different cloud providers to supply their workload traces. This can also be used as a common input format by various scheduling simulators. Our simulator, *GeoSim* which is explained in section VI-A also used this format to represent jobs.

C. Google Workload Trace

Google trace has job and task information in different files. We consolidated all the required information and generated a single work trace file with the format described in the previous section. For our work, we model five different data centers located at different parts of the globe. A detailed explanation about the data center profile are given in section IV. To simulate workload execution in these five data centers we have extracted five set of traces each worth five days of incoming job requests from the original Google trace. This will cover 25 days of traces out of the 29 available days. We label these 5 set of traces from A to E. Additionally, in our experiments we only consider those jobs that end without any failures. Therefore, we clean the original traces by removing all jobs that have either failed or got killed. The left over 4 days of the workload is used to gain statistical user related insights to predict the running time of incoming jobs.

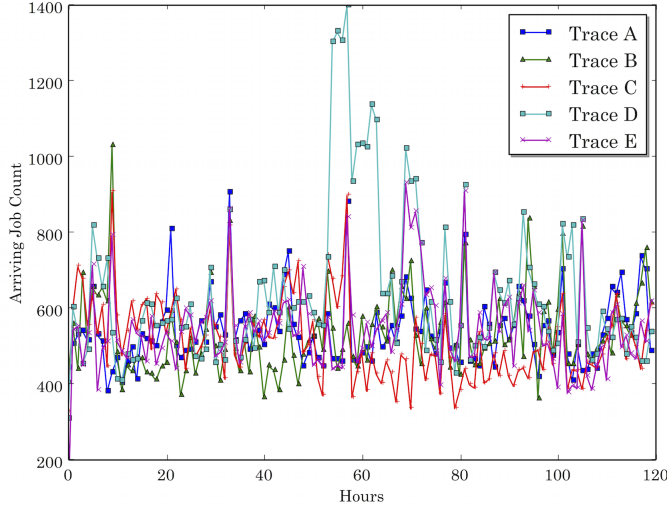


Fig. 1: Arrival rate of jobs

Now we will look at some characteristics of the traces. Table II describes the statistics for each of the five traces grouped by the scheduling class of the job. The CPU and memory usage in Google traces are standardized based on the maximum capacity of the resource available on any machine in the trace, which is assigned a value of 1.0. Most of the class 0 and class 1 jobs are batch jobs with less latency requirements. Generally, these are the kind of jobs that can be provisioned on other data centers as their deadlines are not strict. You will find that as the sensitivity increases the running time of these jobs and the resource usage also increase. High-sensitive jobs are generally service jobs and run of longer periods of time. Figure 2 shows a pie chart of the job counts in different schedule classes for all the 5 traces. You will notice that the job count for class 3 (high latency-sensitive) jobs are low. This is because these jobs generally run for long period of time, typically months. Since we only consider jobs that start and end within the 29 day range such jobs are missed out in our traces. For our work, we will consider class-0 and class-1 jobs as candidates for optimizing the provisioning. Class-2 and class-3 jobs are provisioned to run on the data center where the job arrived since these are high-sensitive and their SLA demands will be strict.

Figure 1 shows there is both intra-trace variations as well as inter-trace variation in job arrival. This is representative of the real-world cloud workload pattern where bursty arrivals are common. There are opportunities to provision jobs to other data centers if the traffic at a particular data center is high. What is also important to notice is that the scheduling decision should be made quickly when the arrival rates of jobs are high, else there are chances for that these jobs miss the SLAs. Hence, even the scheduler should be scalable during high volume traffic.

IV. DATA CENTER PROFILING

In order to run our experiments, we need to collect information about data centers which will help us make better

scheduling decisions. We call this process of collecting the required information about data centers as *data center profiling*. The major energy costs in data center are from running the IT resources and from cooling the data centers. We create a profile for each of the data center with the information that effects the energy costs.

A. Geo-Distributed Data Center

Most cloud service providers including Google, Amazon, Facebook and Apple have data centers distributed across the globe. For our experiments, we consider five locations where Google has its data centers[25]. Table III provides the location and labels for each of the data center. We also replicate the computing resource used in the original Google trace across all the data centers. The configuration of the resources along with the normalized CPU and memory are given in table IV

B. Profile Parameters

We consider two major parameters that has the most effect on the energy cost. They are a) Temperature b) Electricity. We collect temperature and electricity prices for these locations from historical archives. We look at each of these parameters and how they effect energy costs in sections IV-B1 and IV-B2.

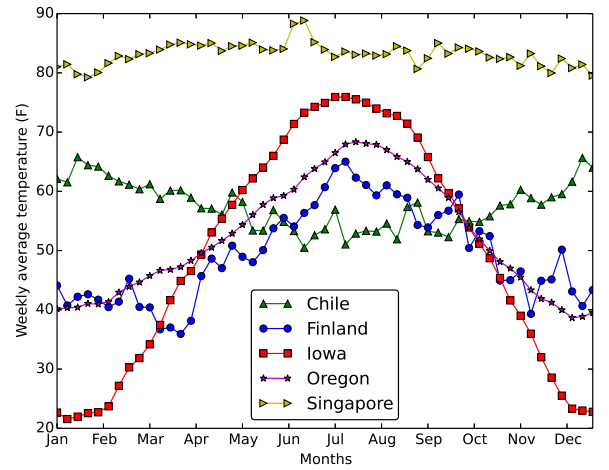


Fig. 3: Temperature variation for the year 2013

1) *Temperature*: Data centers situated in regions with lower temperatures can benefit from utilizing outside air for cooling operations. The use of an air-side economizer enables the distribution of outside air to the servers, while the exhaust air is directed outside instead of being recirculated and cooled. This cost-effective technique can result in significant savings for data centers annually. The cooling costs at different data centers vary due to differences in temperature, which can be leveraged by scheduling jobs at data centers with lower temperatures. To support this approach, we conducted an empirical analysis of historical climate data, sourced from the National Climate Data Center [26] for all five locations as they were in the year 2013. Figure 3 plots the weekly average temperature for five different locations in United States, Europe, South-America, and Asia. As expected we see high diversity in

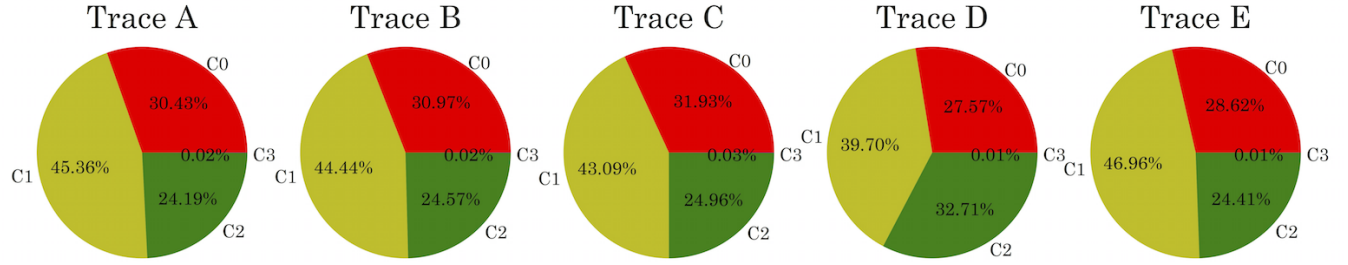


Fig. 2: Distribution of job count in various schedule classes

	Trace A			Trace B			Trace C			Trace D			Trace E		
Sched Class	Avg CPU	Avg Mem	RT (secs)	Avg CPU	Avg Mem	RT (secs)	Avg CPU	Avg Mem	RT (secs)	Avg CPU	Avg Mem	RT (secs)	Avg CPU	Avg Mem	RT (secs)
0	0.075	0.051	415	0.123	0.096	345	0.083	0.065	385	0.093	0.092	373	0.093	0.074	333
1	0.087	0.029	729	0.086	0.032	760	0.082	0.028	711	0.073	0.030	743	0.093	0.043	693
2	0.067	0.043	1028	0.096	0.058	872	0.088	0.057	918	0.073	0.045	707	0.084	0.049	832
3	0.088	0.12	1105	0.099	0.12	82	0.101	0.012	7961	0.081	0.051	17135	0.089	0.061	9021

TABLE II: Trace set statistics

Location	Label	Associated Trace
Council Bluffs, Iowa	Iowa	A
The Dalles, Oregon	Oregon	B
Singapore	Singapore	C
Quilicura, Chile	Chile	D
Hamina, Finland	Finland	E

TABLE III: Data center location information

Node count	Architecture	CPU	Memory
6732	B	0.5	0.50
3863	B	0.5	0.25
1001	B	0.5	0.75
795	C	1	1
126	A	0.25	0.25
52	B	0.5	0.12
5	B	0.5	0.3
5	B	0.5	0.97
3	C	1	0.50
1	B	0.5	0.06

TABLE IV: Data center IT resource configuration

temperature at different locations. For example, Oregon and Finland seems to be equally favorable for cooling during most of the year except during the months of May - July. Chile and Singapore show almost no variation in temperature across the year. However, Chile is favorable for cooling conditions compared to Singapore.

2) *Electricity Pricing*: The establishment of power trading zones has led to the procurement of electricity at wholesale prices based on dynamic pricing models. They are popular in the US [27] and Germany [28], among other countries. An analysis of supply and demand bids displays the possibility of high fluctuations in prices [29]. There are many ways to predict these prices [30] [31]. If we can predict the electricity prices for the job duration, then we will be able to schedule the job such that the running costs go down. Figure 4 shows the variation in electricity prices across different data center regions in 2013. These price values have been consolidated

from multiple sources on the Internet [32] [33] [34]. For Chile, we were able to retrieve only monthly averages for 2013, but for all the other location we were successful in collecting hourly prices. From the figure 4 it is clear that Singapore has the highest electricity pricing among all the regions under consideration. If we compare both figure 3 and figure 4 it is evident that Singapore is least suitable data center in terms of operational costs since the temperature and the energy prices are the highest.

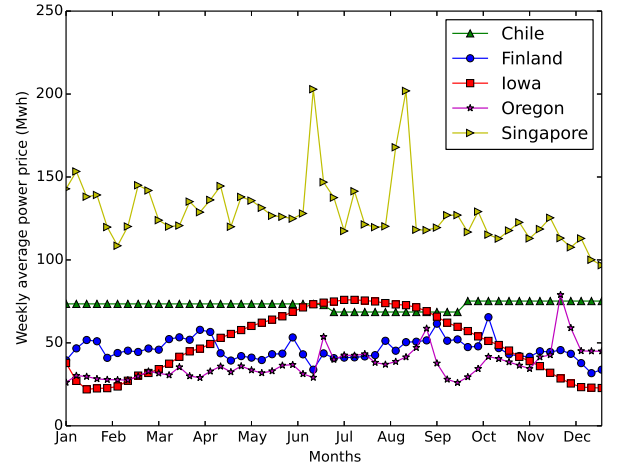


Fig. 4: Electricity price variation for the year 2013

V. SYSTEM MODEL

The energy consumed by a data center can be modeled as the sum of energy used by both its servers (IT equipment) and the cooling subsystem. The subsection V-A and V-B provides more information about the model.

$$E_{total} = E_{server} + E_{cooling} \quad (1)$$

A. Server Energy

Our model considers the total power consumption of servers by aggregating the power consumption of each core within the system. The power consumption of each core is the combination of the static power required to keep it running and the dynamic power necessary to perform computations. Eq 2 represents the power consumed by a core. Static power is consumed when the system is idle.

$$P_{core} = P_{static} + P_{dynamic} \quad (2)$$

The peak usage is when the cores are at the highest utilization. In cloud systems, it is very rare that the CPU utilization u is more than 20% [35]. To account for this, we rewrite eq 2 as described in [36].

$$P_{core}(u) = P_{static} + P_{dynamic_{peak}} * u \quad (3)$$

For our experiments we use the peak and idle power from [37]. The authors have characterized power usage of systems by running various HPC kernels. Since these kernels are CPU-intensive we can consider them to use the peak power. To emulate the CPU utilization of jobs, we generate random utilization based on the *mean*, *standard deviation* and *cov* provided in the work done on data center utilization by IBM [35].

The energy consumption of a job j with core count c , average CPU utilization u and running time t is given in Eq 5

$$P_j = c * P_{core}(u) \quad (4)$$

$$E_j = P_j * t \quad (5)$$

B. Cooling Energy

We model the energy consumption of the traditional CRAC as well as the air economizer in the sections below.

1) *CRAC cooling units*: The power usage of CRAC units can be modeled using the equation 6 as given in [38]. The value of COP (coefficient of performance) can be calculated according to [39] using the equation 7.

$$P_{CRAC} = \frac{P_{dc}}{COP} \quad (6)$$

$$COP = 0.0068 * (T_{sup}^2) + 0.0008 * T_{sup} + 0.458 \quad (7)$$

In the above equations, P_{dc} is the total power consumed by a data center. T_{sup} is the temperature, in degree Celsius, of air supplied to the data center.

Temperature range (F)	PUE
less than 25	1.05
25-35	1.07
35-50	1.09
50-60	1.10
60-65	1.17

TABLE V: PUE values for different temperature ranges

2) *Air Economizer*: The power usage model for air economizer is given in [38]. The equation 8 represents the power consumption of the free air cooler.

$$P_{air} = (PUE - 1) * P_{dc} \quad (8)$$

The PUE(power usage effectiveness) is proportional to the outside temperature. We get the values of PUE from the product documentation of commercial vendors [40]. For our experiments, the value of PUE used is given in table V.

C. Data center selection

The data center is selected by calculating the power contributed by a job to the total server power consumption as well as cooling power. This is done by using Eq 4 in the power models for server and cooling. Equations 9, 10 and 11 show the mechanism of data center selection.

$$P_{total}(j, dc) = P_{dynamic}(j, dc) + P_{cooling}(j, dc) \quad (9)$$

$$Cost(j, dc) = T_j * P_{total}(j, dc) * C_{mwh}(dc) \quad (10)$$

$$dc_j = \min_{dc \in DC} Cost(j, dc) \quad (11)$$

The cost is the product of total power due to job j in a specific data center dc , the job running time $T(j)$ and the cost of electricity $C_{mwh}(dc)$. The job scheduling process involves selecting the data center that has the minimum cost associated with it as specified in Eq 11.

VI. EVALUATION

A. Simulator

We evaluate our meta-scheduler by doing extensive simulations. For this purpose we developed a simulator *GeoSim* which take in the workload trace and the data center profile as input and implements our model and routes jobs to the most cost effective data center. A brief overview of the design of *GeoSim* is provided below. We have made the source code of *GeoSim* and the workload traces public and is available at [41]

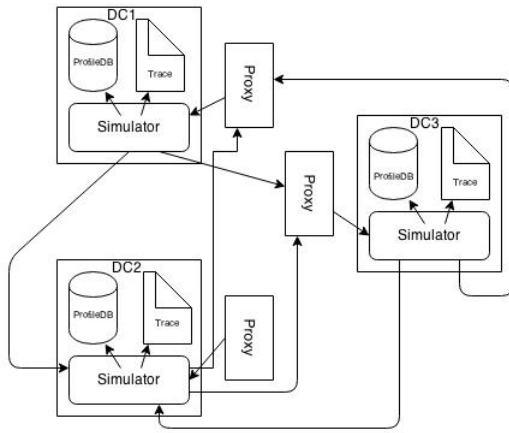


Fig. 5: Simulation components and their interaction

1) Components:

- **Datacenter:** This component will simulate a data center. This composed of 1) scheduler that implements a specific provisioning algorithm, 2) data center profile database, 3) a work trace reader and 4) proxy elements of other data center for inter-data center communication
- **DataCenter Proxy:** All interactions with the data center will be through a proxy. Each data center will hold the proxies to other data centers.
- **Scheduler:** This component will implement the model and schedule jobs onto data centers. We can experiment with different scheduling algorithms by plugging different Scheduler implementation
- **Datacenter Profile DB:** This components contains the temperature and electricity data. Scheduler will access this to make provisioning decisions.

2) *Component Interaction:* Figure 5 gives the interaction model between the components. The data center object will start a thread which will run the scheduling algorithm, scheduling jobs from the trace file. Each data center has proxies to other data centers if it needs to communicate resource availability or job provisioning to other data centers. In our simulations we share resource utilization every 5 minutes, and each time step of the simulator is 5 seconds. These are provided as parameters to the simulator. Our simulations take around 4 hours to complete running on 6-core AMD Opteron(tm) server.

	Energy (MJ)	Cost (\$)
DumbSched	8044	560630
GeoSched	7384	494560
Savings!	660	66070

TABLE VI: Result summary

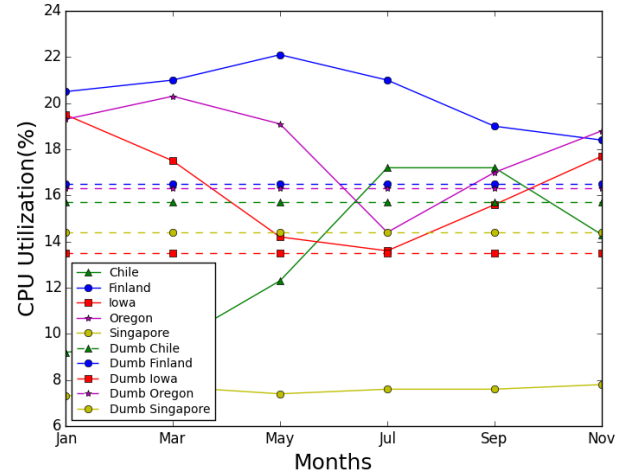


Fig. 6: Data center utilization

B. Results

We run the simulations with GeoSched and also implement a naive scheduler called *DumbSched* which schedules all the jobs to the local data centers. We will look at the cost saving as well as the energy savings with GeoSched. We will also look at the variation in utilization of each of the data centers across different months. We run the simulations for alternate months starting with January. We provide the date 15 and GMT time 12am as the start date and time for each simulation.

Table VI shows our findings. The amount of energy consumed by data centers for 30 days (5 days in each of the 6 months) and the corresponding costs are shown. The energy decrease by 8.2% and the cost decrease by 11.7%. When these results are extrapolated to 1 year the dollars saved is around \$4.8 million. The experiments were run assuming a max core count in node to be 64. Since, Google traces normalize these values, it would be impossible to calculate energy with out an assumption.

Figure 6 shows the variation of data center utilization when the scheduling month changes. It is quiet evident that Singapore has the least utilization due to high operational costs. Similarly, the utilization at Iowa decreases towards may and increase towards the end of the year. This result is consistent with the temperature and electricity price data. Finland and Oregon has the highest utilization since these two have less operational cost as compared to the other three data centers. Chile, gets more jobs towards the middle of the year. This could be attributed to temperature increase in other data centers.

Figure 7 shows how energy consumption varies. Note that some data centers consume more energy with GeoSched than

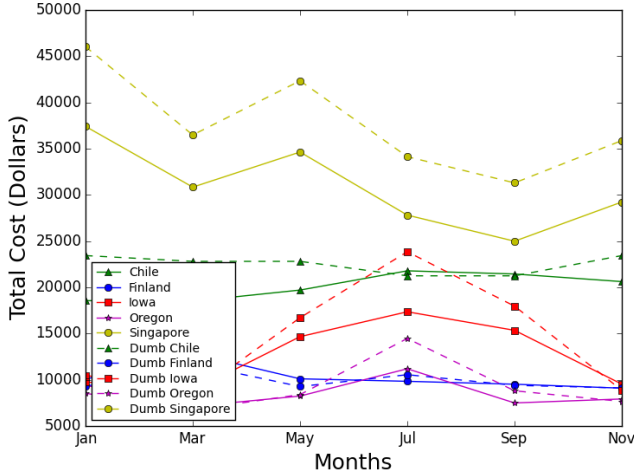


Fig. 8: Variation in operational cost

DumbSched. This is because, in these data centers, more work can be done with less increase in energy consumption. Therefore, GeoSched will give a net decrease in energy consumption even if some data centers consume more energy than when scheduled using DumbSched.

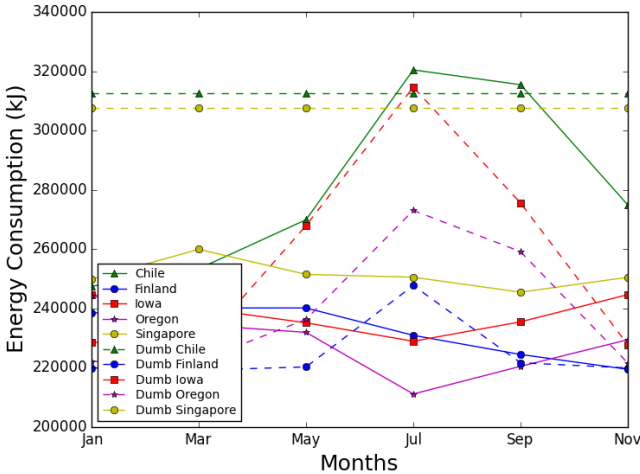


Fig. 7: Variation in energy consumption

Figure 8 shows the cost of operation among different data centers. This plot can also be interpreted the same way as the energy consumption plot. The use of GeoSched results in a reduction of the overall cost of operating the data center. Further energy savings can be achieved by switching off the unused nodes within the data center. To account for this, the idle energy consumption of these machines is calculated in all of our experiments.

VII. CONCLUSION

In this paper, we build a meta-scheduler for geo-distributed data centers. We considered parameters like temperature and

electricity prices to intelligently provision jobs to data centers. We evaluate our method using simulations and found our scheduler to reduce energy consumed by 8.2% and the cost of operation by 11.7%. For future work, we aim to investigate additional factors that affect energy consumption, such as the availability of renewable energy sources and the energy required for data transfer. We also intend to develop more comprehensive Service Level Agreements (SLAs) to incorporate into our scheduling framework. Additionally, we plan to examine how workload diversity affects our meta-scheduling approach, since different types of workloads have varying requirements and priorities that must be accommodated. Furthermore, we plan to explore the scalability of our framework for larger and more intricate data center networks, which would require more sophisticated resource management strategies and algorithms.

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